Band-1 Hadamard order 14428 – Classical Paley-II DIRECT (q2=7213); IN SPK via Paley-II; Gap 2052 Still OPEN

H_14428 Paley-II PASS; Paley-I FAIL/SKIP (q=14427=3^2 7229 NOT prime); Kronecker FAIL/SKIP (4*3607; 3607 odd prime NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-09

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-10-09_Band1_Hadamard_14356_PaleyII` (Paley-II construction + chain template); `2026-10-09_Band1_Hadamard_14424_PaleyI` (current bookkeeping); `2026-10-09_Band1_Hadamard_14420_Paley` `2026-10-09_Band1_Hadamard_14412_PaleyI`; `2026-10-09_Band1_Hadamard_14408_PaleyI`; `2026-10-09_Band1_Hadamard_14388_PaleyI`; `2026-09-21_Band1_Hadamard_9124_PaleyII`; ....
Lab. `/workspace/experiments/band1/hadamard_14428_paleyII/`.
Cutoff. 2026-10-09T23:10:00Z (~7:10PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite construction of order 14428 only (verified max|HH^T-nI|=0), via classical Paley-II q2=7213 (prime ===1 mod 4), order 2(q2+1). 14428 IN SPK via Paley-II. 14428 is a long-known Hadamard order; this is a finite on-box verification, NOT a new-order claim.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 14428: Paley-II **PASS** ($q_2 = 7213$ prime $\equiv 1 \pmod 4$, $n = 2(q_2 + 1)$); Paley-I **FAIL/SKIP** ($q = 14427 = 3^2 \cdot 7 \cdot 229$ NOT prime); Kronecker **FAIL/SKIP** ($14428 = 4 \cdot 3607$; 3607 odd prime, NOT Hadamard-admissible; do NOT invent H_3607). **14428** folds into SPK **only** via classical Paley-II (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED** (no invented J). No finished-ToE claim. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **14424+14420+14412+14408+14388+14356+14348+14328+14324+14304+14252+14244+14208+141** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

Abstract

Finite construction of order **14428** (second Paley-II DIRECT wave of the 14xxx chain, after 14356).

- H_{14428} = Paley-II ($q_2 = 7213$) – classical SPK
- Paley-I for 14428: **FAIL/SKIP** ($q = 14427 = 3^2 \cdot 7 \cdot 229$ NOT prime)
- Kronecker for 14428: **FAIL/SKIP** ($4 \cdot 3607$; 3607 odd prime, NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_14428.py`

Results. overall_pass=True; units 789/789; row blocks 113/113 (block size 128); wall 2492.1875s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-09T23:10:00Z (~7:10PM ET)

Arithmetic

- Target $n = 14428 = 2(q_2 + 1)$ with $q_2 = 7213$ prime $\equiv 1 \pmod 4$ — Paley-II eligible (claim path).
- Paley-I: $q = n - 1 = 14427 = 3^2 \cdot 7 \cdot 229$ NOT prime — FAIL/SKIP.
- Kronecker: $14428 = 4 \cdot 3607$; 3607 odd prime ($\equiv 3 \pmod 4$) NOT Hadamard-admissible — FAIL/SKIP; no other admissible split.
- Shape: $14428 = 2^2 \cdot 3607 \equiv 12 \pmod{16}$ (exact n/m shape twin 13996, unbuilt; template 14356 was $\equiv 4 \pmod{16}$ with odd composite m).
- Miyamoto digression skipped this wave (prefer classical SPK).

Construction (Paley type II)

Let χ be the quadratic character of $\mathbb{F}_{7213}$ and $Q_{ij} = \chi(j - i)$ (symmetric, since $q_2 \equiv 1 \pmod 4$). The conference matrix $C = \begin{pmatrix} 0 & \mathbf{1}^T \\ \mathbf{1} & Q \end{pmatrix}$ of order $q_2 + 1 = 7214$ gives

$$H_{14428} = \begin{pmatrix} C + I & C - I \\ C - I & -(C + I) \end{pmatrix}.$$

Built directly in int8 (memory-lean; unit-tested equal to the int64 reference builder on $q \in \{5, 13, 17, 29, 37, 41, 53, 61\}$), then verified block-wise: every row block of $HH^T - nI$ checked exactly in int32.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_14428 Paley-II	Paley-II q2=7213	0	2473.659
H_14428 Paley-I	FAIL/SKIP (q=14427=3ˆ27229 NOT prime)	–	–
H_14428 Kronecker	FAIL/SKIP (4*3607; 3607 odd prime NOT Hadamard-admissible)	–	–

SPK closure note

Quantity	Value
14428 in SPK	**True** (Paley-II q2=7213)
Prior 14424	IN SPK via Paley-I (~6:05PM ET 2026-10-09)
Prior Paley-II 14356	IN SPK via Paley-II q2=7177 (~1:22PM ET 2026-10-09)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED (no invented J)
- No finished-ToE claim
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- 14428 is a long-known order (Paley 1933 family); finite on-box verification only
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

NEXT direct route after 14428: q=14431->n=14432 Paley-I DIRECT (q=14431 prime ===3 mod 4; no prime in 14428..14430; Paley-II q2=7215=3*5*13*37 NOT prime power -> FAIL/SKIP; Kronecker 14432=4*3608; 3608=2ˆ3*11*41 ===0 mod 4 admissible with H_3608 via Paley-I q=3607

-> ALSO classical, note-only, NOT on box (do NOT double-construct); 14432=2?*11*41 ===0 mod 16; template hadamard_14424_paleyI) | then Paley-I q=14447->n=14448 (no direct route in 14436..14444) | FIRST Paley-II DIRECT after 14428: n=14460 (q2=7229 prime ===1 mod 4; Paley-I q=14459=19*761 NOT prime -> FAIL; Kronecker 14460=4*3615; 3615=3*5*241 odd NOT Hadamard-admissible -> FAIL/SKIP; template hadamard_14428_paleyII) | no Kronecker-only route in 14432..14460 (sympy scan) | skipped Paley-II DIRECT orders below 14428 still unbuilt (each Paley-I FAIL, Kronecker odd-m FAIL/SKIP; template hadamard_14428_paleyII / hadamard_14356_paleyII): 13924 (q2=6961) / 13956 (q2=6977) / 13996 (q2=6997; exact 14428 n/m shape twin 2^2*3499) / 14004 (q2=7001) / 14028 (q2=7013) / 14116 (q2=7057) / 14140 (q2=7069) / 14220 (q2=7109) / 14260 (q2=7129) (+13900 q2=6949 historically not listed)